

1) Find the eigenvalues of  $A$ , given  $A = \begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}$

$$A\vec{v} = \lambda\vec{v} \rightarrow (A - \lambda I)\vec{v} = 0, \vec{v} \neq 0 \quad A - \lambda I = 0$$

$$\begin{bmatrix} 6-\lambda & 0 \\ 0 & 4-\lambda \end{bmatrix} = A - \lambda I$$

$$\det(A - \lambda I) = 0 \rightarrow (6-\lambda)(4-\lambda) - 0 = 0$$

$$\lambda = 6, 4$$

2) Two corresponding eigenvectors are

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ and } b \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \forall a, b \in \mathbb{R}$$

3) Given  $A\vec{v} = 7\vec{v}$ , what is  $A^5\vec{v}$ ?

$$A^5\vec{v} = 7^5\vec{v}$$

4) Why must  $\det(A - \lambda I) = 0$ ?

Because that's when  $A\vec{v} = \lambda\vec{v}$ , that is to say the transformation  $A$  is equal to the scalar product. It leaves the direction fixed, and only changes the vector's magnitude.

5) Suppose a covariance matrix has eigenvalues

100, 12, 3, 0.4. Without doing any calculations, What does this tell you about the underlying data? What would you expect Principal Component Analysis to do?

The covariance matrix is 4-dimensional, so the data tracks 4 features.

The principal component 1 accounts for the around 90% of the data, with the second accounting for around 10% of the data. The other two account for very little of the data.

PCA would tell us the directions for which the data is most correlated.